
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and that local density removes most of it at $d = 20$, half at $d = 25$
8 and little of the already small $d = 32$ value. A second-order expansion of a linear
9 map restricted to the manifold explains why: a readout sees the second fundamen-
10 tal form contracted with its normal component, $\langle w_N, \Pi \rangle$, and, at second order, an
11 alignment’s error splits into a connection mismatch and the relative second funda-
12 mental form projected into the target normal space; mean curvature retains only
13 the metric trace of Π and determines neither. On a unit-normalized embedding
14 that Hessian splits into a sphere term proportional to the readout’s value and a
15 shape term. On a synthetic surface the sphere term holds at -0.74 to -0.80 across
16 samplings that swing the $\|H^S\|$ partial from -0.29 to $+0.26$; on 86,471 labelled
17 galaxies the mismatch between the label Hessian and the full probe-facing Hessian
18 is negative wherever the label has local structure and survives cross-fitting and a
19 change of encoder, the sphere term is negative on two of four labels under a weak-
20 ridge sensitivity, and the shape term alone takes a label-dependent sign consistent
21 with that decomposition. Mean curvature of a single representation is therefore
22 not a viable explanation for weak Platonic convergence; the obstruction is relative
23 and, for the fitted global linear alignment, already present at first order.

24 1 Introduction

25 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
26 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
27 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
28 a family of different vision models share only 0.3 to 2.3 percent of their nearest neighbours across
29 the two surveys. We ask where on the manifold that agreement lives: if two models disagree most
30 where the embedding surface bends sharply, local mean curvature should associate negatively with
31 local cross-survey agreement.

32 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
33 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
34 test of the curvature–alignment prediction, which finds no effect free of density; (iii) the residual
35 expansion of a linear map on a submanifold, which names the curvature such a map responds to and

shows mean-curvature magnitude is blind to it; and (iv) known-surface and real-data tests on the readout side, where the expansion’s terms behave as predicted.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. Embeddings are unit-normalized (data on S^{767}). Curvature is evaluated only where J has full rank.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape. To isolate shape, the map differentiated is $F/\|F\|$, whose H splits exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}); H^S , normal to M within the sphere, carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder. Decoder curvature has served as a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is a validated instrument.

Validation. Scored against closed-form curvature on rank (Spearman), direction (cosine) and magnitude ratio. At $D = 768$ an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$ (equivalently $\|H\|$, the radial part being fixed), direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local, correlated with $\|H_{\text{raw}}\|$ over all 10,000 points at $d \in \{20, 25, 32\}$.

Result. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density (the $k = 30$ nearest-neighbour estimate on the Legacy embedding, $d = 20$) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 : 78%, 49% and 8% of the association removed. Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; three decoder seeds; the CKA [Kornblith et al., 2019] tertile gap within strata agrees at $d = 20$ and 25). Removing the radial component of H_{raw} (the raw surface is off the sphere; the two rank at $\rho = 0.89$ – 0.96) collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$, so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

To second order, a linear function is as non-linear on M as M bends in that function’s normal direction [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the

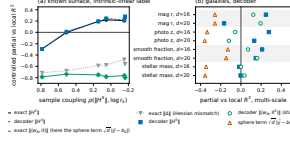


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample’s density–curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder’s $\|H^S\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

rest, $\Pi = \Pi^S - g \hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaussian constant; real k -NN patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced connections agree at x_0 (in particular under a local intrinsic isometry); the extrinsic term’s normal part is $\Pi_G - P_N^G A \Pi_F$.

So $\|H\|$, the norm of the metric trace of Π , need not correlate with either $\langle w_N, \Pi \rangle$ or the relative term; the second-order term scales as s^4 and, with fixed- k neighbourhoods, density sets s , so curvature effects scored on patches are density-sensitive; and when the label has intrinsic curvature of its own, Δ carries the residual and the curvature piece can take either sign.

5 The probe-facing curvature, on a known surface and on galaxies

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman–Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at $k \in \{16, \dots, 2048\}$ (five scales). Per anchor we compute $\|H^S\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface is resampled with weight $\|H^S\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H^S\|$ partial follows the sampling from -0.29 to $+0.22$, and the decoder reproduces it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ has partial correlation -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2) (rank 0.996–1.000 with $\sqrt{d} |\hat{y} - b_0|$).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares

quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours (a local Monge chart, where the Christoffel symbols vanish at the anchor, so the quadratic coefficient is the covariant Hessian). Three regularities (Table 1; other decoder seeds and widths in Appendix A). The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass, where the sealed-probe sign came with ridge shrinkage of extreme predictions. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). Writing $K = \langle w_N, \Pi \rangle$, curvature lowers the second-order residual relative to $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$; positive alignment is the first-order condition (bending toward the label’s curvature helps until it overshoots); the mismatch column is the full test. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift), consistent with Eq. (3): it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so raising it shrinks or enlarges the mismatch according to that condition; empirically its sign follows the cross term’s where that is nonzero and the remaining square where it is null. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55–0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace on five encoders at the same anchors (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$, $k = 2,048$) has density-controlled associations with the magnitude probe’s local R^2 of -0.24 , 0.00 , $+0.27$, $+0.23$ and -0.19 : a trace statistic has no encoder-stable sign, consistent with Eq. (1). The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A decoder yields a pointwise second fundamental form at production scale that agrees with an independent estimator and with the data. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the Hessian mismatch carries the predicted negative sign, robust to cross-fitting, probe strength and the choice of encoder; the sphere term does so on two of four labels under a weak-ridge sensitivity; and the shape term is label-signed because it enters through its alignment with, and magnitude relative to, the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. If weak convergence is attributed to the failure of a linear alignment map, one embedding’s mean curvature is not the relevant second-order quantity; after tangent alignment the relevant normal term is $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot ($d = 20$, two seeds; Appendix C) finds the fitted global ridge map carries HSC tangent frames into the Legacy embedding with 44% residual, a first-order mismatch, and no estimator-stable second-order association (three estimators disagree in sign). On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10–0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces (-0.14 to -0.88) but the alignment partial has no stable sign: alignment is a regularity of the galaxy data, not a demonstrated mechanism. Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

References

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://>

- 164 [//arxiv.org/abs/2405.07987](https://arxiv.org/abs/2405.07987).
- 165 Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation
166 models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025.
167 URL <https://arxiv.org/abs/2509.19453>.
- 168 Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv*
169 *preprint*, 2025. URL <https://arxiv.org/abs/2508.10104>.
- 170 Hiroaki Aihara, Nobuo Arimoto, Robert Armstrong, et al. The Hyper Suprime-Cam SSP Survey: Overview
171 and survey design. *Publications of the Astronomical Society of Japan*, 70(SP1):S4, 2018. doi: 10.1093/pasj/
172 psx066.
- 173 Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI Legacy Imaging Surveys. *The Astro-*
174 *nomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.
- 175 Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large obser-
176 vation models for astronomy. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2405.14930>.
- 177 Yonghyeon Lee and Frank Chongwoo Park. On explicit curvature regularization in deep generative models. In
178 *2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, ICML, 2023.
179 URL <https://arxiv.org/abs/2309.10237>.
- 180 Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic
181 curvature in neural manifolds. *arXiv preprint*, 2022. URL <https://arxiv.org/abs/2212.10414>.
- 182 Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors
183 in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531, 2010.
- 184 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network rep-
185 resentations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*,
186 2019.
- 187 John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer,
188 2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.
- 189 Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.
190 arXiv:1610.01644.
- 191 David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of*
192 *Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.

193 **Appendix A: decoder ablations**

194 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
195 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
196 variance explained 0.952). Every significant sign of the main table is retained.

197 **Appendix B: sensitivity of the mismatch and sphere columns**

198 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local
199 R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no
200 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak
201 ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which
202 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that
203 probe.

label	quantity	seed 0	seed 1	seed 2	400 ³
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	−0.38	−0.34	+0.33	+0.31	+0.35	−0.21 (−0.26)
mag_r	20	−0.43	−0.40	+0.39	+0.37	+0.24	−0.22 (−0.26)
photo_z	16	−0.42	−0.36	+0.23	+0.20	+0.32	+0.05* (−0.33)
photo_z	20	−0.42	−0.42	+0.22	+0.22	+0.20	+0.03* (−0.36)
smooth_fraction	16	−0.10	−0.17	+0.17	+0.19	+0.33	−0.12 (−0.13)
smooth_fraction	20	−0.15	−0.21	+0.05*	+0.03*	+0.23	−0.13 (−0.15)
stellar_mass	16	+0.01*	−0.04*	−0.03*	+0.06*	+0.31	−0.02* (−0.35)
stellar_mass	20	−0.06*	−0.07*	+0.01*	−0.00*	+0.19	−0.02* (−0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder’s $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L \cdot, L \cdot) - P_N^G A \Pi_F\|$, the same with L fitted on 256 neighbours’ latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F ’s tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	−0.15	+0.04*	−0.18	+0.04*	−0.18	+0.13
$d = 25$, seed 0	0.64	0.43	−0.10	+0.06	−0.16	+0.05	−0.23	+0.13
$d = 20$, seed 1	0.64	0.44	−0.11	+0.05	−0.19	+0.04	−0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$,

217 multi-scale density control. The first row per label is the probe’s global out-of-sample R^2 ; Table 6
218 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per
encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	−0.11	+0.16	+0.16	−0.21
	sphere	−0.26	−0.23	−0.40	−0.24	−0.08*
	mismatch	−0.39	−0.57	−0.34	−0.32	−0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	−0.06*	−0.05*	−0.17	+0.29	−0.16
	sphere	−0.33	−0.21	−0.29	−0.46	−0.44
	mismatch	−0.45	−0.61	−0.45	−0.48	−0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	−0.08*	+0.20	+0.24	+0.06*
	sphere	−0.13	−0.06*	+0.05*	−0.15	−0.11
	mismatch	−0.09*	−0.33	−0.25	−0.20	−0.01*
	alignment	+0.19	−0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	−0.15	−0.10	−0.11	−0.13
	sphere	−0.35	−0.26	−0.38	−0.40	−0.38
	mismatch	−0.04*	+0.03*	−0.12	−0.06*	−0.02*
	alignment	+0.01*	−0.05*	−0.01*	+0.07*	−0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.